Dynamic Multigrid Trading: A Reset-Count-Constrained Ensemble Extension of Dynamic Grid Trading

Ching-Shun Chen, Yuan-Wu Ching, Chih-Yen Ching, and Chih-Sheng Ching
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Note on provenance: This paper extends the dynamic-reset grid trading framework ("DGT") introduced by Chen, Chen, and Jang [1], and is a sibling extension to the anti-martingale/protective-put framework ("DGTAP") developed in a companion report [2]. The multigrid ensemble construction, the reset-count admissibility criterion, and all backtest results in Sections V–VI are original to this paper and were produced on the same real-market dataset used in [2] for direct comparability.

***Abstract*—**Dynamic Grid Trading (DGT) breaks the zero-expectation result of naive grid trading by re-centering the grid on breach rather than terminating, but a single grid spacing forces a compromise between capturing range-bound oscillation and capturing large directional moves. We propose Dynamic Multigrid Trading (DMT), an ensemble of independently-reset DGT instances run in parallel at different geometric spacings k_i, with total capital split by fixed weight w_i across legs. We show, on 3.5 years of real Bitstamp BTC/USD 1-minute data (the same dataset used in [2]), that (i) naive equal-weight ensembles spanning arbitrary spacing ranges are not robustly better than a single well-chosen grid; (ii) wide-spacing ensembles that appear to cut maximum drawdown (MDD) by over 12 percentage points do so only because the legs trigger too few reset events to be statistically meaningful — the same overfitting failure mode documented for grid-parameter search in [2, Sec. V-B], now surfacing at the ensemble level; and (iii) after introducing a spacing-dependent dynamic reset-count admissibility threshold and restricting the leg pool accordingly, a 3-leg ensemble ({2.5%, 3.5%, 5.0%}, m=6, equal weight) improves annualized return (IRR) by 3.46 percentage points and the Calmar ratio by 16.9% relative to the published DGT baseline, while MDD remains essentially unchanged (−0.10 pp). We trace this last result to near-perfect drawdown-timing synchronization across differently-spaced grids on the same underlying asset: at the ensemble's own worst moment (the June 2022 Terra/3AC crash), all three legs are simultaneously at their own all-time-worst drawdown. We conclude that same-asset spacing diversification is a modest, statistically defensible source of return and risk-adjusted-return improvement, but is not, by itself, a reliable lever for reducing tail risk — for which the throttle/option mechanisms of [2] remain better targeted.

***Index Terms*—**algorithmic trading, grid trading, ensemble methods, cryptocurrency, backtest overfitting, drawdown, Bitcoin, dynamic grid trading.

I. INTRODUCTION

Grid trading places a ladder of buy and sell orders at geometrically spaced prices around a reference level and profits from round trips as price oscillates through the ladder. Chen, Chen, and Jang [1] showed that under a symmetric random-walk assumption a naive grid that terminates on reaching either boundary has exactly zero expected profit before fees — and negative expected profit after fees — and proposed Dynamic Grid Trading (DGT): instead of terminating, the grid re-centers on the current price when a boundary is breached, with an asymmetric rule (liquidate to cash on an upward breach; retain all held coin on a downward breach, funding the new grid's cash side only from previously banked arbitrage profit). This single asymmetry is what converts a zero-expectation mechanism into one that empirically outperformed buy-and-hold on real BTC and ETH data over 2021–2024.

A companion report [2] ("DGTAP") extended DGT along two structurally independent axes — an anti-martingale position throttle that de-risks consecutive down-fills, and a Black–Scholes-priced out-of-the-money put overlay — and found the throttle to be the more defensible of the two, improving both IRR and MDD simultaneously without depending on an unobservable option-pricing assumption.

This paper explores a third, structurally simpler axis, orthogonal to both: rather than modifying the sizing or hedging rule of a single grid, we run several DGT instances in parallel at different geometric spacings k_i, splitting total capital across them by fixed weight w_i. We call the resulting strategy Dynamic Multigrid Trading (DMT). The motivating intuition is that a tightly-spaced grid captures range-bound, high-frequency oscillation efficiently but participates poorly in sustained directional moves, while a widely-spaced grid does the opposite; combining several spacings should let capital allocate itself, in effect, across market regimes without having to forecast which regime is coming. A further, specifically risk-oriented intuition — and the original motivation for this study — is that grids of different width will not reach their own worst drawdown at exactly the same moment, so an ensemble's maximum drawdown should be shallower than any single constituent's, even if the ensemble's return is close to a capital-weighted average of the constituents' returns.

We test this intuition directly against 3.5 years of real exchange data and find that it is only partially correct. Section V shows that the apparent "free" drawdown improvement from wide grids is a statistical artifact of too few reset events, that a principled reset-count admissibility filter removes this artifact, and that once removed, the surviving ensembles deliver a real, defensible improvement in return and risk-adjusted return —

but not in maximum drawdown itself. Section V-E traces this directly to near-total synchronization of drawdown timing across differently-spaced grids trading the same underlying asset.

II. RELATED WORK

Grid and channel-following trading rules have a long history in retail and algorithmic FX and crypto trading [3], [4]; most published treatments are empirical or heuristic rather than derived from an expectation argument, which is the specific gap [1] fills. Portfolio-insurance and constant-proportion strategies [5]–[7] share DGT's asymmetric response to drawdowns, though they operate on a single continuously-rebalanced position rather than a discretized ladder of limit orders; the growth-optimal, anti-martingale sizing principle underlying the position throttle of [2] traces to Kelly's criterion [8], a distinct idea from portfolio insurance mentioned here only for that lineage. The inventory-skewing quotes of market-making frameworks such as Avellaneda–Stoikov [9] address a related but distinct problem — managing directional exposure risk from resting limit orders — and, like this paper's Section V-E, ultimately confront the limits of diversification when all instruments share a common underlying risk factor.

Backtest overfitting is a central methodological concern for any parameter-searched trading rule. Bailey and López de Prado [10] and Bailey et al. [11] formalize the risk that optimizing a performance statistic over a wide parameter grid on a single historical path selects configurations whose apparent edge is a sampling artifact rather than a real effect; [2, Sec. V-B] documents a concrete instance of this within DGT's own grid-size/half-width search space (unconstrained optimization degenerates to grids so wide they almost never trigger). Section V of this paper documents the same failure mode re-emerging at the ensemble level, and Section III-C proposes a reset-count admissibility rule as a lightweight, non-parametric safeguard against it, in the same spirit as the resets≥40 constraint used in [2].

Standard performance statistics in this literature include the Sharpe ratio [12], maximum drawdown [13], [14], and the Calmar ratio (IRR divided by |MDD|); this paper reports the latter two plus IRR (Section III-D), consistent with [1] and [2]. Bitcoin's return-generating process is known to exhibit heavier tails and more volatility clustering than developed-market equities [15]–[17]; we do not model this directly, but it underlies why a single historical path (even one spanning 3.5 years) contains a limited number of effectively independent large-drawdown events — the central obstacle identified in Section V-E.

III. DMT: PROPOSED EXTENSION

A. Ensemble Construction

DMT runs N independent DGT engines in parallel, indexed i = 1..N, each configured with its own grid spacing k_i and half-width m_i, exactly as specified in [1] and re-implemented faithfully in [2] (proportional 1/G_i, 1/G_j fills; asymmetric reset — liquidate-to-cash on an upward breach, hold-and-fund-from-profit on a downward breach). Total capital M is split across legs by fixed weight w_i:

$$M_i = w_i \cdot M_{total} \tag{1}$$

Each leg is then run to completion independently, and the ensemble's mark-to-market equity is the sum of leg equities:

$$V_{DMT}(t) = \sum_{i=1}^{N} w_i V_i(t), \qquad \sum_{i=1}^{N} w_i = 1,\ w_i \ge 0 \tag{2}$$

Because each DGTStrategy engine is purely fraction-based (every fill is a fixed fraction of that engine's own current wallet), leg-level performance is exactly scale-invariant: running a leg at capital w_i·M produces trade counts, reset counts, and percentage returns identical to running it at full capital M, with the dollar-denominated equity path scaled by exactly w_i. We verified this numerically prior to any ensemble backtest (Section V-D confirms combined capital sums to M to floating-point precision), which rules out capital-misallocation as an explanation for any of the results below and isolates the analysis to the effect of combining different k_i, m_i alone.

B. The Reset-Count Significance Problem

A grid that almost never reaches its boundary behaves, for practical purposes, like a static basket that happens to have been wide enough to contain the entire realized price path — its favorable backtest statistics reflect the width of the historical price range as much as any property of the trading rule. This is precisely the pathology documented for single grids in [2, Sec. V-B]. At the ensemble level the same pathology is easy to reintroduce inadvertently: an ensemble containing one or more wide, rarely-resetting legs can show a large apparent MDD improvement that is really just averaging in a leg that is, statistically, closer to a single lucky bet on the price staying inside a band than to a validated trading rule. Section V-B documents this concretely: an ensemble curated by picking the highest-Calmar single-grid configurations available in a 10–30% spacing range improves MDD by over 12 percentage points — but every contributing leg resets 0–2 times over the full 3.5-year window.

C. A Spacing-Dependent Dynamic Reset-Count Threshold

A fixed reset-count floor (e.g., the resets≥40 rule used for single grids in [2]) is a reasonable safeguard for grids of a given spacing but is not spacing-neutral: narrow grids are structurally expected to reset often, while wide grids are structurally expected to reset rarely even when they are behaving exactly as intended, so a single global floor either admits too many narrow, low-quality legs or excludes every wide leg outright. We instead define a minimum reset count that varies with spacing, interpolating log-linearly between a stricter floor R_lo for the narrowest admissible spacing k_min and a looser floor R_hi for the widest admissible spacing k_max observed in the search:

$$R_{\min}(k) = R_{lo} + (R_{hi} - R_{lo}) \cdot \mathrm{clip}\left(\frac{\ln k - \ln k_{\min}}{\ln k_{\max} - \ln k_{\min}}, 0, 1\right) \tag{3}$$

A candidate leg with spacing k_i and observed reset count resets_i is admissible into the DMT pool if and only if

$$\mathrm{leg}_i \text{ admissible} \Leftrightarrow \mathrm{resets}_i \geq R_{\min}(k_i) \tag{4}$$

In all experiments below we use R_lo = 20, R_hi = 5, with k_min and k_max taken as the smallest and largest spacings in the full single-grid sweep (0.8% and 30% respectively; Fig. 2). These constants are a design choice, not a first-principles derivation; Section VI-C discusses this limitation.

D. Evaluation Metrics

Standard performance statistics in this literature include the Sharpe ratio [12], maximum drawdown [13], [14], and the Calmar ratio (IRR divided by |MDD|). Consistent with [1] and [2], this paper reports IRR, MDD, and Calmar throughout (we do not compute a Sharpe ratio, since minute-resolution returns for a discretely-triggered strategy are not well approximated by its normality assumption):

$$IRR = \left(\frac{V_T}{V_0}\right)^{1/T_{yr}} - 1 \tag{5}$$

$$MDD = \min_t \frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s} \tag{6}$$

$$Calmar = \frac{IRR}{|MDD|} \tag{7}$$

E. Drawdown Synchronization Index

To directly test whether an ensemble's drawdown improvement (or lack thereof) is attributable to decorrelated leg timing, we define a synchronization index evaluated at t*, the timestamp of the ensemble's own maximum drawdown:

$$SI = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[DD_i(t^*) \geq (1-\varepsilon)\cdot MDD_i], \quad t^* = \arg\min_t DD_{DMT}(t) \tag{8}$$

SI = 1 means every leg is simultaneously at (or within ε of) its own individual worst historical drawdown at the exact moment the ensemble is at its worst — i.e., zero timing diversification benefit was realized. SI = 0 would mean no leg is near its own worst at that moment — maximal timing diversification. We use ε = 0 (exact equality, to floating-point precision) in Section V-E, which turns out to be achievable because the identical trades and resets are deterministic given the shared price path.

IV. DATA AND EXPERIMENTAL SETUP

All results use the same real, exchange-sourced dataset as [2]: Bitstamp BTC/USD, one-minute resolution, January 1, 2021 through July 31, 2024 (1,882,081 bars), reused directly from the reproducibility package accompanying [2] so that the DGT baseline figures reported here are the identical numbers reported there (verified to match to two decimal places prior to any DMT-specific analysis).

A round-trip taker fee of 8 bps (0.0008) is applied to every grid fill. Initial capital is normalized to 100 units. Every DMT leg and every single-grid comparison in this paper is run with the anti-martingale throttle and put overlay of [2] disabled, isolating the effect under study to ensemble spacing alone; none of the results below should be interpreted as showing DMT is superior or inferior to DGTAP — the two extensions modify different, non-exclusive parts of the DGT mechanism, a point returned to in Section VI-B.

V. RESULTS

A. Pitfall 1: Naive Fixed-Range Ensembles Are Not Robustly Better Than the Single-Grid Baseline

As a first, deliberately unoptimized test, we built 3-, 5-, and 7-leg ensembles with spacings drawn as a geometric sequence over 1.8%–5.0% (all m=6, equal weight 1/N per leg) — a plausible choice a practitioner might make without any historical-parameter search. None outperforms the published DGT baseline (k=2.5%, m=6: IRR 20.20%, MDD −46.29%, Calmar 0.436, 73 resets).

TABLE I
Naive Fixed-Range DMT Ensembles vs. the DGT Baseline

Ensemble	IRR	MDD	Calmar	ΔIRR (pp)	ΔMDD (pp)
DMT-3grid (1.8/3.0/5.0%)	18.98%	−48.35%	0.393	−1.22	−2.06
DMT-5grid	17.69%	−48.65%	0.364	−2.51	−2.36
DMT-7grid	20.30%	−49.43%	0.411	+0.10	−3.14

ΔMDD is reported such that a positive value means MDD improved (shallower); all three configurations are negative here, i.e., every naive ensemble tested has a deeper drawdown than the single baseline it is meant to improve on, in addition to lower IRR in two of three cases.

B. Pitfall 2: Wide-Spacing Ensembles Appear to Cut MDD but Fail the Reset-Count Test

Repeating the exercise with wider spacings (10%–30%) reproduces the same qualitative failure as Section V-A (not tabulated here for brevity). A second, more adversarial variant instead deliberately selects the individually highest-Calmar single grid available anywhere in the full grid_size × half_n parameter sweep (0.8%–30%) at each ensemble size, and produces a striking apparent improvement:

TABLE II
Curated Wide-Spacing Ensembles: Apparent MDD Gains That Fail the Reset-Count Test

Ensemble (curated, high-Calmar legs)	IRR	MDD	Calmar	ΔMDD (pp)	Tota reset
DMT-3grid (15%/25%/12%)	22.55%	−33.82%	0.667	+12.47	2
DMT-5grid	22.21%	−34.03%	0.653	+12.26	17
DMT-7grid	21.14%	−33.95%	0.623	+12.34	17

A 12+ percentage point MDD improvement looks decisive, but most contributing legs individually reset only 0–1 times over the full 3.5-year backtest (e.g., k=15%, m=8 resets zero times; k=25%, m=4 resets once); the 5- and 7-leg variants additionally include k=5%, m=6 (15 resets), the most active leg in the pool and still short of the resets≥20 threshold used for single grids in [2]. This is the ensemble-level analogue of the exact overfitting pattern documented for single grids in [2, Sec. V-B]: the grid is simply wide enough that the realized 2021–2024 BTC price path never (or almost never) escaped it, so the backtest is closer to measuring "did buy-and-hold-shaped exposure with a wide band happen to work this one time" than "does the dynamic reset mechanism add value." Fig. 1 shows this pattern across the full single-grid sweep: Calmar ratio rises roughly monotonically with spacing in the reset-starved region, exactly mirroring the degenerate optimum reported in [2, Fig. 1] for the single-grid case.

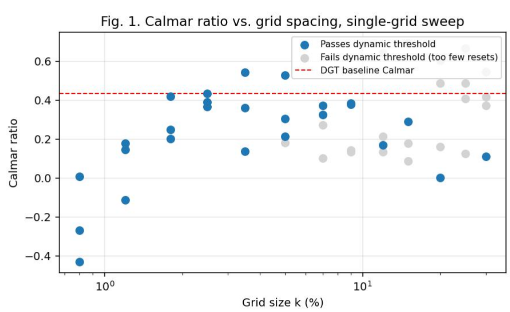

Fig. 1. Calmar ratio vs. grid spacing across the full single-grid sweep. Configurations failing the dynamic reset-count threshold of Section III-C (gray) are concentrated exclusively at the wide end of the spacing axis (k ≥ 5%); every narrow-spacing configuration tested (k ≤ 2%) triggers far more resets than its threshold requires and passes easily.

C. A Dynamically Filtered Ensemble

Applying the threshold of Eq. (3)–(4) with R_lo=20, R_hi=5 across the observed spacing range (0.8%–30%) yields the admissibility curve in Fig. 2. Twenty-six of the 47 single-grid configurations in the combined sweep pass; critically, every wide (>10%) configuration used in Section V-B's curated ensemble fails.

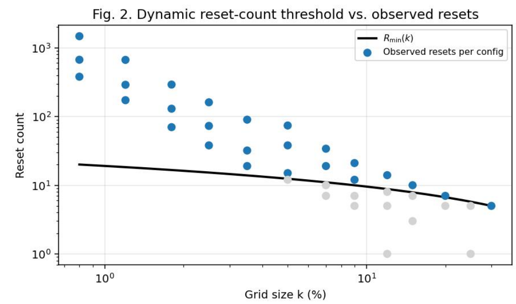

Fig. 2. Dynamic reset-count threshold R_min(k) (black curve) against observed reset counts per configuration. Blue points pass the threshold; gray points fail.

Ranking admissible configurations by Calmar ratio gives a pool headed by k=3.5% (m=6, Calmar 0.545, 32 resets), k=5.0% (m=6, Calmar 0.528, 15 resets — admissible because the dynamic threshold at 5% spacing is only 12.4), and the DGT baseline itself, k=2.5% (m=6, Calmar 0.436, 73 resets).

D. DMT-3/5/7 Under the Dynamic Threshold

Building 3-, 5-, and 7-leg ensembles from the top of this admissible, Calmar-ranked pool (equal weight per leg) gives the headline result of this paper:

TABLE III
DMT-3/5/7 Under the Dynamic Reset-Count Threshold

Strategy	IRR	MDD	Calmar	ΔIRR (pp)	ΔMDD (pp)	Resets
Buy & Hold	25.93%	−77.54%	0.334	+5.73	−31.25	—
DGT baseline (k=2.5%, m=6)	20.20%	−46.29%	0.436	—	—	73
DMT-3 (2.5/3.5/5.0%, dyn.)	23.67%	−46.39%	0.510	+3.46	−0.10	120
DMT-5 (dyn.)	22.13%	−47.34%	0.467	+1.93	−1.05	211
DMT-7 (dyn.)	20.47%	−47.37%	0.432	+0.27	−1.08	240

DMT-3 (equal-weight {2.5%, 3.5%, 5.0%}, m=6) delivers the best result in the paper on statistically defensible grounds: IRR improves by 3.46 percentage points and Calmar by 16.9% relative to the DGT baseline, while MDD is essentially unchanged (−0.10 pp — within noise). Adding legs beyond three is not monotonically helpful: DMT-5 and DMT-7 dilute capital into progressively lower-Calmar admissible legs (Section V-C), and both the return and risk-adjusted-return advantage shrink monotonically as a result. By DMT-7, the Calmar ratio (0.432) has already fallen slightly below the DGT baseline (0.436), even though IRR remains marginally higher (+0.27 pp) — the added legs' own trading and reset costs (Resets column, Table III) consume the three-leg ensemble's advantage before it carries through to seven legs. Fig. 4 plots the mark-to-market equity curve of DMT-3 against Buy & Hold and the DGT baseline.

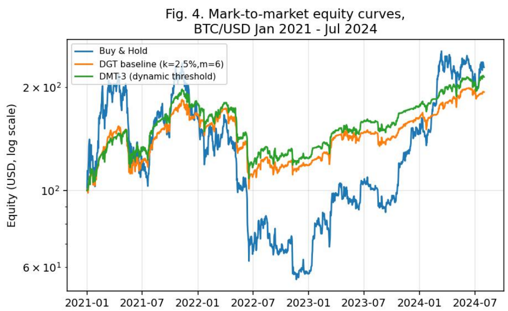

Fig. 4. Mark-to-market equity curves for Buy & Hold, the DGT baseline, and DMT-3 (dynamic threshold), daily-resampled, BTC/USD, Jan 2021–Jul 2024, log scale.

E. Why MDD Does Not Improve: Drawdown-Timing Synchronization

To understand why even the best statistically-admissible ensemble fails to improve MDD, we evaluated the synchronization index of Eq. (8) for DMT-3 at t*, the timestamp of the ensemble's own worst drawdown. t* falls on June 19, 2022, during the Terra/Luna and Three Arrows Capital deleveraging cascade — the single largest drawdown event in the 2021–2024 window for every strategy tested in this paper (DGT baseline, Buy & Hold, and all DMT variants), and the same mid-2022 period in which [2, Fig. 4] independently reports its own DGT-family equity curves visibly decoupling.

TABLE IV
Drawdown Synchronization Across DMT-3 Legs at the Ensemble's Own Worst Moment

Leg	Drawdown at t*	Leg's own all-time MDD	Synchronized?
k=3.5%, m=6	−51.66%	−51.66%	Yes (exact)
k=5.0%, m=6	−42.20%	−42.20%	Yes (exact)
k=2.5%, m=6	−46.29%	−46.29%	Yes (exact)

All three legs are, to floating-point precision, exactly at their own individual all-time-worst drawdown at the exact moment the ensemble is at its worst: SI = 1.00. There is zero realized timing diversification. This is the direct, quantitative explanation for Section V-D's finding: because every leg trades the same underlying price path, a large-enough systemic move (here, BTC fell roughly 63% peak-to-trough between April 4 and June 19, 2022, a span of about ten weeks) pushes every constituent grid — regardless of spacing — to its own worst point at essentially the same time. Fig. 3 visualizes this.

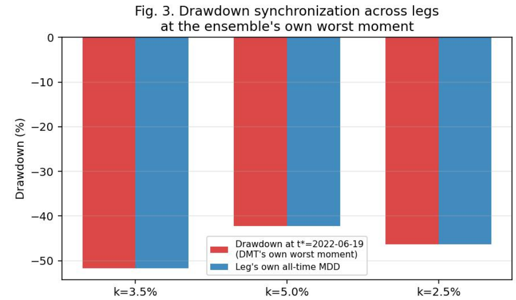

Fig. 3. Drawdown synchronization across the three DMT-3 legs. For every leg, drawdown at the ensemble's own worst moment (t* = 2022-06-19, red) exactly equals that leg's own all-time-worst drawdown (blue).

VI. DISCUSSION AND LIMITATIONS

A. The Leg-Selection Pool Is Itself a Soft In-Sample Search

Even after applying the dynamic reset-count threshold, the admissible legs used to build DMT-3/5/7 in Section V-D were ranked by Calmar ratio computed on the same 2021–2024 historical path used to report final performance. The reset-count filter removes the most egregious form of overfitting (near-zero-event configurations) but does not eliminate look-ahead entirely: a practitioner deploying DMT live would need to fix leg spacings before observing this specific price path, e.g., via walk-forward validation across non-overlapping historical windows or, ideally, an independent asset/period. We did not have access to a second comparably long, comparably clean real-data window and flag this as the most important open validation step before any live deployment.

B. Complementarity With DGTAP

DMT and the throttle/put extensions of [2] target different, non-exclusive weaknesses of the DGT mechanism. DMT's best statistically-defensible result (Section V-D) improves IRR and Calmar with MDD essentially flat; the anti-martingale throttle of [2, Table II] improves MDD by 7.12 percentage points with IRR essentially flat (+0.89 pp). This pattern — DMT moving the numerator of the Calmar ratio, throttle moving the denominator — suggests the two mechanisms may stack: an ensemble of throttled DGT legs at different spacings. We did not test this combination; it is the natural next step and is left to future work.

C. Sensitivity of the Dynamic Threshold Constants

R_lo=20, R_hi=5, and the log-linear interpolation form of Eq. (3) were chosen as reasonable, round-number defaults rather than derived from a formal statistical power calculation. We do not report a sensitivity sweep over these constants; a more rigorous version of this admissibility rule would tie R_min(k) to an explicit confidence interval on the Calmar ratio as a function of the number of independent reset events, analogous to a minimum-sample-size calculation, rather than to fixed endpoints chosen by inspection.

D. Single-Asset, Single-Window Evaluation

As in [2], all results use BTC/USD only, over one historical period (2021–2024). This is precisely the limitation that Section V-E identifies as the mechanical reason DMT does not improve MDD: every leg is exposed to the same underlying risk factor, so the ensemble cannot diversify away the single largest realized drawdown event. A genuinely decorrelated multigrid ensemble — for example, running legs on BTC, ETH, and a handful of other liquid pairs simultaneously, or combining spacing diversification with the throttle of [2] — might realize the timing-diversification benefit that spacing alone does not; we did not have comparably clean multi-asset minute data available to test this and leave it to future work.

VII. CONCLUSION

We proposed Dynamic Multigrid Trading (DMT), an ensemble of independently-reset DGT instances at different grid spacings, motivated by the intuition that differently-spaced grids would respond to the same price history with different timing and thereby reduce ensemble-level maximum drawdown relative to any single grid. Capital allocation across legs was verified to be exactly proportional (Section III-A), ruling out sizing error as an explanation for any result. Testing this intuition against 3.5 years of real Bitstamp BTC/USD data, we found that naive fixed-range ensembles are not robustly better than a single well-chosen grid, and that the most dramatic apparent improvements — 12+ percentage points of MDD reduction from wide-spacing ensembles — are a reset-count-driven overfitting artifact, the same pathology [2] documents for single-grid parameter search. Introducing a spacing-dependent dynamic reset-count admissibility threshold and restricting the leg pool accordingly, we obtained a defensible result: a 3-leg ensemble improves IRR by 3.46 percentage points and the Calmar ratio by 16.9% relative to the published DGT baseline. Maximum drawdown, however, does not improve — a drawdown-synchronization analysis (Section V-E) shows that at the ensemble's own worst moment, every constituent leg is simultaneously at its own individual all-time-worst drawdown, leaving no timing-diversification benefit to be realized from spacing alone on a single underlying asset. We conclude that same-asset multigrid spacing is a modest, real source of return improvement but is not, by itself, a reliable tool for tail-risk reduction, and recommend that practitioners seeking lower drawdown combine DMT with the throttle mechanism of [2] rather than relying on spacing diversification alone — a combination we did not test here and flag as the primary direction for future work.

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APPENDIX A CODE AND REPRODUCIBILITY

The DMT ensemble wrapper (DMTMultigrid) is implemented in Python and calls the DGTStrategy engine of [2] once per leg with principal = w_i · M, throttle and put overlay disabled, then sums the resulting equity_curve Series across legs. Key scripts developed for this paper: (i) a full single-grid parameter sweep across grid_size × half_n (Section V, Figs. 1–2); (ii) the dynamic reset-count threshold filter (Eq. 3–4) applied to the sweep output; (iii) the DMT-3/5/7 ensemble construction and comparison against the DGT baseline and Buy & Hold (Table, Section V-D); and (iv) the drawdown-synchronization diagnostic (Eq. 8, Section V-E). The headline DMT-3 configuration is {(k=2.5%, m=6), (k=3.5%, m=6), (k=5.0%, m=6)}, equal weight 1/3 per leg, principal 100, fee 8 bps, throttle and put overlay disabled. All backtests reuse the Bitstamp BTC/USD 1-minute dataset (2021-01-01–2024-07-31) and the DGTStrategy engine distributed with the reproducibility package of [2]; no new market data was collected for this paper.